Glossary

abdomen
The abdomen is one of three main body segments of insects (which also include the head and thorax). The abdomen contains the digestive and reproductive systems.

altruism
Altruism, biologically speaking, is animal behavior that benefits another animal at the expense of the first, in terms of survival or reproduction. Worker bees that forgo their own reproduction in order to help the queen reproduce is one of the most extreme examples of altruism in nature. See kin selection.

amino acid
Amino acids are the building blocks of proteins. They are a necessary part of bees' (and humans') diets, necessary to function and develop. Pollen is the source of amino acids in bee bread and royal jelly.

anther
The anther is the part of a flower that contains pollen.

apiary
An apiary is a collection of one or more hives, also called a bee yard. Most backyard beekeepers have only one apiary, but commercial beekeepers may manage dozens.

Apidae
Apidae is a family classification of bees. It is the largest family of bees and includes honey bees, bumble bees, carpenter bees, and many others.

Apis
Apis is a genus classification of bees. It includes all bees that produce and store honey.

Apis mellifera
Apis mellifera is the scientific name for the Western honey bee. *Apis* is the genus and *mellifera* is the species. *Mellifera* means "bearer or maker of honey" (*melli* means "honey" and *fera* means "bearer").

arthropods (Arthropoda)
Arthropoda is one of the many phyla of the animal kingdom. Honey bees are in the phylum Arthropoda, and are therefore known as arthropods. All arthropods are invertebrates and have jointed bodies and paired jointed appendages.

bee bread

Bee bread is pollen that has been moistened by honey and honey bee glandular secretions and fermented by microorganisms such as yeast, bacteria, and fungi. Pollen in the form of bee bread is honey bees' major source of protein, fats, and vitamins, and is an important nutrient source for the development of brood.

bee space

Bee space is 3/8", which leaves enough room for bees to pass one another on opposite combs. If the space between combs is less than 3/8", bees will likely fill it with propolis. It if is greater than 3/8", the bees will tend to build burr comb to fill the space.

bee yard

See apiary.

beeswax

See wax.

brood

Brood is the name given to the egg, larvae, and pupae stages of insects.

brood nest

The brood nest is the part of the hive where the brood is developing, distinct from the area where honey is being processed and stored. Conditions such as temperature are more important here, for the healthy development of brood.

brood pheromone

Brood pheromone is released by larvae and indicates its sex, caste, and age. It influences development and behavior of other bees, most notably by inhibiting the development of ovaries in workers.

burr comb

Burr comb is comb built anywhere the beekeeper doesn't want it, namely between frames or hive bodies. It is likely to be built in spaces with dimensions that exceed the width of bee space.

carbohydrate

Carbohydrates are the sugars in food that are a source of energy for organisms.

caste

A caste is one form within a species that may differ from other forms in terms of anatomy, physiology, and reproduction. The three honey bee castes are queens, workers, and drones.

chromosome

Chromosomes are the structures containing an organism's genetic material in the form of DNA.

class

Class is a rank in the classification of life, below phylum and above order. Honey bees are members of the class Insecta (insects).

cleansing flights

A cleansing flight is the delicate name given to the act of leaving the hive to defecate. Bees can delay defecation for months during winter, but will leave the hive on a warm day to void their waste. Fecal staining may be evident around or on the outside of hives in winter and early spring.

cluster

(also winter cluster)
Clustering is a thermoregulatory behavior where bees congregate on the comb to keep warm when environmental temperatures are below 64°F. Colony functions such as foraging and water collection cease when bees are clustering.

cocoon

A cocoon is a silky pupal casing spun by insect larvae to protect them during their pupal stage.

colony

The colony is the family community of honey bees that share a nest. There is usually a queen present, many workers, and fewer drones.

comb

see honeycomb

Complementary Sex Determination gene

The Complementary Sex Determination gene is responsible for sex determination in honey bees.

compound eye

The two large eyes found on either side of a bee's head. Each compound eye consists of thousands of individual independent facets with lenses that collectively form a coherent image.

cooperative brood care

Cooperative brood care is the practice of sharing the work of raising young that is not one's own. In the case of honey bees, the queen lays the eggs, but the larvae, pupae, and young adults are cooperatively cared for by their older sisters, nurse bees. Cooperative brood care is one of the three hallmarks of eusociality.

corbicula
(plural: corbiculae)
See pollen basket.

crop
See honey crop.

diploid organism
Diploid organisms have paired sets of chromosomes in each cell (excluding sperm and egg cells). They inherit one set of chromosomes from their mother and one from their father. Female honey bees (both queens and workers) are diploid, because they develop from eggs that are fertilized (by sperm) and therefore contain two sets of chromosomes; in contrast, drones are haploid. See also haplodiploid.

DNA (deoxyribonucleic acid)
DNA is the long molecule that carries an organism's complete set of genetic information.

drift
Drift is the entering of bees into hives or nests that are not their homes. It is more likely when hives are close together and difficult to distinguish from one another. Hives at the end of a row of hives may contain a lot of drifting bees.

drone
A drone is a male honey bee.

drone comb
Drone comb is specialized brood cells where unfertilized eggs are laid and reared into drones. The cell size is slightly–but obviously–bigger than that of worker comb and it is often built on the margins of frames.

drone congregation area
A drone congregation area is where hundreds of drones congregate high in the air to wait for a virgin queen on her mating flight.

Dufour's gland
The Dufour's gland is located at the end of the abdomen in female bees and releases pheromones through the vagina. In the queen, these pheromones have a similar role to queen mandibular pheromone, serving as a signal of her fertility and promoting retinue behavior. When a colony loses its queen, laying workers may start producing their own queen-like fertility pheromones from their Dufour's glands.

dysentery

Dysentery is honey bee diarrhea, which may be visible inside or outside of the hive. It is caused by the build-up of solids in the digestive tract, and can relate to infections.

egg

Eggs are the first stage of the bee's life, and are laid by queens and sometimes workers into cells. The pattern of egg-laying in the cells and across the comb can provide information about the health and trajectory of the colony.

emergency queen cells

Emergency queen cells are built when workers abruptly find themselves without a queen and must convert larvae that had been destined to be workers into queens instead.

eusociality

(adjective: eusocial)
Eusociality refers to the most advanced form of social living. Eusocial individuals fully depend on a societal organization to survive and cannot live on their own. The defining characteristics of a eusocial organism are more than one generation living together, cooperative brood care, and a reproductive division of labor.

exoskeleton

An exoskeleton is an organism's outer body shell, made of tough, protective material. All invertebrates have an exoskeleton and do not develop a backbone.

family

Family is a rank in the classification of life, below order and above genus. Honey bees are members of the family Apidae.

fat body

The fat body is a large organ distributed throughout an insect's body and typically surrounding the gut and reproductive organs. Its main function is to store fat and glycogen to release energy for use in times of food shortage. Winter bees have larger and more numerous fat bodies than summer bees.

festooning

Festooning is when inactive bees hang between combs, linking themselves together by their legs. It is associated with wax construction.

field bees

See foragers.

forage

Foraging is the search for and collection of resources (nectar, pollen, water, and resin) for the colony.

foragers

Foragers (also known as field bees) are worker bees that take foraging trips for nectar, pollen, water, or resin. Typically, worker bees begin foraging halfway through their lives, and remain foragers until they die. Foragers specialize in collecting one kind of resource, but they can be flexible as needed. See also house bees and nurse bees.

ganglia

(singular: ganglion)

Ganglia are nerve centers in the thorax and abdomen that constitute the nervous system. In the honey bee, ganglia control muscle movements more than the brain does.

gland

Glands are honey bees' organs that secrete chemical substances. See Dufour's gland, hypopharyngeal gland, mandibular gland, Nasonov gland, and wax gland.

genome

A genome is an organism's complete set of genetic material carried by DNA. The genome includes information for a variety of elements, the most important being genes.

genes

Genes are the working subunits of DNA and are the basic unit of heredity, passed down to an individual's offspring through reproduction. Genes act as instructions to make proteins and other molecules.

genus

Genus is a rank in the classification of life, below family and above species. Honey bees are members of the genus *Apis*. Genus names are always italicized.

hamuli

(singular: hamulus)

Hamuli are hooks that connect the pair of wings on each side of a bee while it is flying, giving the appearance of honey bees having only two wings instead of four.

Haplodiploid organisms

(noun: haplodiploidy)

Honey bees as a species are haplodiploid. Males, or drones, are haploid organisms, meaning each of their cells possesses a single set of chromosomes inherited from their mother. Females are diploid organisms, meaning each of their cells (excluding sperm and eggs) possesses two sets of chromosomes (one inherited from their mother and one from their father).

Haploid organisms

Male honey bees are haploid organisms, meaning each of their cells possesses a single set of chromosomes inherited from their mother. See haplodiploid.

heater bees

Heater bees generate heat in the brood nest to ensure that pupae are incubated at the proper temperature (91-97°F, or 33-36°C).

hemolymph

Hemolymph is analogous to blood in the honey bee. It circulates freely through the bees' bodies, rather than being confined to arteries, capillaries, and veins as in the human body.

hindgut

The honey bee hindgut is the posterior part of the digestive system where waste is stored for eventual excretion through the rectum. It also contains waste filtered from the hemolymph by Malpighian tubules.

hive

The hive is a physical, manmade structure that houses a colony of bees and their nest of brood and food. The word "hive" is often mistakenly used interchangeably with "colony", but it should be distinguished between the bees and their physical home.

honey

Honey is the sugary food substance that honey bees produce from concentrating and chemically converting nectar. Water is evaporated from the nectar until the water content is approximately 17-19%, and enzymes and bacteria are added to break down sucrose in nectar into fructose and glucose. When this process is complete, the nectar has been converted into honey. The bees cap the cells of honey with wax for storage.

honeycomb

See comb.

honey bound

(alternate spelling: honeybound)
A honey bound colony has nectar and honey filling the cells in the brood nest, restricting the area in which the queen can lay eggs.

honey crop

The honey crop (also called a honey stomach) is a specialized part of the gut that stores nectar, honey, or water. A valve called the proventriculus at the base of the crop prevents nectar or water that a forager has collected from moving into the ventriculus and being digested.

honey flow
See nectar flow.

honey stomach
See honey crop.

hormones
Hormones are chemicals secreted into hemolymph and carried to specific organs and tissues to influence how they function. Unlike pheromones, hormones do not leave the body. In the honey bee, as in humans, hormones are involved in reproduction, development, and the general functioning of the body.

house bees
House bees are worker bees that perform necessary tasks inside the hive, such as tending to brood or the queen (when they are known as nurse bees, or the queen's retinue, respectively).

hygienic behavior
Hygienic behavior is a trait where workers are able to detect, uncap, chew down and/or remove infected pupae. It is a mechanism of social immunity to help reduce the spread of pathogens.

Hymenoptera
Hymenoptera is an order classification of insects. Hymenoptera includes ants, bees, and wasps. They typically have four membranous wings joined together by small hooks (hamuli). They are also the biggest-brained insects. Within Hymenoptera, there are nine families of bees.

hypopharyngeal glands
Hypopharyngeal glands are located in the heads of nurse bees where they produce and secrete royal jelly to feed larvae.

inbred (inbreeding)
Inbreeding is the production of offspring from closely related individuals. An inbred population has a limited number of genes; the probability of deleterious traits that are typically rare usually rise. For this reason, an inbred population may die out when faced with new selection pressures.

insect
Insects are invertebrate animals within the class Insecta. They have three main body parts (head, thorax, and abdomen), three pairs of legs, and one or two pairs of wings.

invertebrates

Invertebrates do not develop a backbone and instead have an exoskeleton, or an outer body shell made of tough, protective material. All arthropods are invertebrates.

juvenile hormone

Juvenile hormone is a hormone found in all insects and it plays an important role in honey bee caste determination and worker polyethism.

kin selection

Kin selection is the evolutionary strategy that favors the reproductive success of an individual's relatives at the cost to the individual's own reproduction and survival.

kingdom

Kingdom is the highest rank in the classification of life. Life classifies into five kingdoms: animals, plants, fungi, bacteria, and archaea. Honey bees are members of the animal kingdom.

larva

(plural: larvae)
The larva is the developmental phase of an insect after it has hatched from an egg but before it has become a pupa. A larva's main function is to eat and grow. All honey bee castes spend an average of six days as larvae.

laying worker

A laying worker is a worker honey bee that lays eggs. This can happen when the queen has died or is no longer producing high levels of queen pheromone. There are sometimes laying workers in queenright hives as well. Laying workers' offspring are drones because their eggs are not fertilized.

light polarization

See polarized light.

Malpighian tubules

Malpighian tubules are slender tube-shaped organs that comprise part of the excretory system in insects. In honey bees, approximately one hundred Malpighian tubules filter waste from the hemolymph and drain it into the hindgut. Their function is analogous to that of kidneys in humans.

mandibles

Mandibles are jaws. Foragers use their mandibles to collect pollen, and honey bees defending their hive may use their mandibles to bite insect invaders.

mandibular gland

The mandibular glands are located near the mouth and secrete pheromones. In the queen, they secrete queen pheromone; in workers, they secrete an alarm pheromone.

mating flight

When a virgin queen is sexually mature, she takes 1-3 mating flights to a drone congregation area far from her colony, where she mates with an average of twelve drones. Mating flights (and swarms) are the only time a queen leaves the hive.

mating sign

A mating sign is the tip of a drone's penis that remains in the queen after they have mated. When a drone mates, ejaculation ruptures his penis and he falls away and dies. The mating sign of the last drone may remain in the queen when she returns to her colony.

Metamorphosis

Metamorphosis is the conspicuous transformation of an organism's body structure from an immature stage (a larva) to an adult stage (an adult bee).

midgut

The midgut of the honey bee is a portion of the digestive system that contains the ventriculus (analogous to a human stomach), where food is digested and nutrients are absorbed.

mushroom body

The mushroom body is the part of the bee's brain in which sensory information is processed, interpreted, and remembered.

Nasonov gland

The Nasonov gland is an organ located near the end of a worker's abdomen that produces and secretes Nasonov pheromone.

Nasonov pheromone

Nasonov pheromone is released by worker bees to orient and aggregate their nestmates. They use it to mark the nest entry to assist returning bees, aggregate during swarming, and mark a water source.

nectar

Nectar is a carbohydrate-rich liquid produced by plants. Bees collect nectar from flowers and turn it into honey. There is great variation in the quality and quantity of nectar in different plants and in the same plants at different times.

nectar flow

(also known as a honey flow)

A nectar flow is a regionally specific time period when particular plants are flowering and secreting a large quantity of nectar, e.g., basswood trees or goldenrod.

neonicotinoids

Neonicotinoids are a class of insecticides, and are the most widely used insecticides in the world.

nurse bee

Nurse bees are young house bees that feed and tend to brood.

ocelli

(singular: ocellus)
The ocelli are three eyes arranged in a triangle pattern near the top of a bees' head that can detect light intensity. Ocelli cannot form an image.

olfaction

Olfaction is the sense of smell. The ability to smell the complex odors of flowers and pheromones is critical to honey bee communication, reproduction, behavior, and foraging.

order

Order is rank in the classification of life, below class and above family. Honey bees are members of the order Hymenoptera.

orientation flight

Honey bees' first journeys from the nest are brief orientation flights to learn their immediate surroundings and nest location. Such flights may also be necessary after a hive has been moved to a new location.

ovarioles

The ovarioles are the tube-like tissues where eggs are produced in the queen's ovaries. Each of the queen's two ovaries is comprised of 150-180 ovarioles, while each of the worker's ovaries have only 2-12 ovarioles.

ovary

Ovaries are a pair of reproductive glands in which eggs are produced. Queens' ovaries are much larger and contain much more ovarioles than workers'.

oviduct

The oviduct is the tube through which an egg passes from an ovary into the vagina.

ovipositor

For most insects, the ovipositor is an egg-laying organ. In honey bees and other Hymenoptera insects, however, it has adapted to become a stinger. Because of this, only female bees can sting.

pathogen

A pathogen is a microorganism (bacteria, virus, etc.) that can cause disease.

pheromone

A pheromone is a chemical produced by specialized glands secreted outside of the body to elicit a behavioral or physiological response by another individual of the same species
(in contrast to hormones that are secreted inside the body). Pheromones can be detected through smell or taste. Pheromone production differs from caste to caste and changes over the course of a bee's lifetime.

phylum

(plural: phyla)
Phylum is a rank in the classification of life, below kingdom and above order. Honey bees are members of the phylum Arthropoda.

polarized light

Polarized light is made of electromagnetic waves that are all vibrating in the same direction. Although humans cannot see it without aid, the sun produces polarized light (even through clouds), which bees can see and use to navigate while foraging.

pollen basket

(also known as corbicula)
Pollen baskets (also known as corbiculae) are on the hind legs of honey bees. A forager uses the pollen combs on her legs to comb off pollen stuck to her body hairs and pack it into a colorful pellet in the pollen baskets. Pollen baskets can also be packed with resin to make propolis.

pollen comb

Pollen combs are structures on worker bees' legs used to move pollen stuck to foragers' body hairs and pack it into a colorful pellet in their pollen baskets.

pollen

Pollen granules are the male gametes of a plant, analogous to sperm in animals. Pollen is an important food source for bees and provides them with protein, vitamins, minerals, and fats. Foragers generally collect pollen from a single kind of flower at a time, which promotes effective flower pollination, but overall the hive collects pollen from multiple kinds of flowers.

polyandry

(adjective: polyandrous)
Polyandry is the practice of females mating with multiple males. Honey bees are polyandrous, as queens mate with an average of twelve males. Polyandry helps promote genetic diversity in colonies.

polyethism

Polyethism is the transitioning of workers through many different tasks as they age. Workers typically spending the first half of their lives as house bees performing various in-hive tasks, and the second half as foragers outside the hive. The shifts in duties are regulated by hormones and pheromones, and are affected by the age of the bee and the social environment of the colony.

proboscis

(plural: proboscises)
A proboscis is the tubular mouthparts of the honey bee (formed when a bee brings together several components of her tongue to form an airtight tube) used to suck up nectar, honey, and water. The proboscis is also used for trophallaxis between bees. It is tucked up behind the chin when it is not needed.

protein

A protein is an organic compound comprised of amino acids. Proteins perform a variety of functions in living organisms to help them develop and function. Proteins are acquired through the diet to provide essential amino acids that cannot be synthesized by the body. Honey bees meet their protein nutritional needs through pollen.

propolis

(verb: to propolize)
Propolis is a sticky substance created and used by bees to caulk and seal holes around the perimeter of their nest and between combs, protect the nest from moisture, and even cover or wall off unwelcome objects. Propolis also provides the colony with antimicrobial protection. Propolis is made from tree resins that foragers carry home in their pollen baskets, mixed with wax. Some subspecies of honey bees tend to produce more propolis than others.

propolis envelope

A propolis envelope is the name for a layer of propolis that sometimes lines the interior of the nest, protecting the colony.

proventriculus

The proventriculus is a valve at the base of the honey crop. When closed, it prevents the nectar that a bee has collected from moving into the ventriculus; when open, it allows food to pass through and be digested.

pupa

(plural: pupae; verb: to pupate)
The pupa is the developmental phase of a honey bee following its stage as a larva and before it emerges as an adult. As a pupa, the bee is undergoing metamorphosis. Honey bee pupae develop within a cocoon in a cell capped with wax.

pupate

See pupa.

queen

A queen is a female bee with a fully developed reproductive system. She is generally mother to all the bees in the colony; indeed, there is typically no more than one queen in a colony at a time.

queen cell

Queen cells have female eggs laid in them that are intended to develop into queens. They may be supersedure cells or emergency queen cells. See also queen cup.

queen cup

Queen cups are cells built by worker bees for raising brood to become queens. They are present in the nest at most times, but are most abundant during the spring swarm season. They tend to be located toward the bottom of frames/comb and oriented downward. They are called queen cells once eggs have been laid in them.

queen pheromone

(Also known as queen substance)
Queen pheromone is the name for the complete blend of pheromones secreted by the queen. It is comprised of nine different components that are released from several different glands on her body, including the mandibular gland. This complex of pheromones attracts drones during mating flights and attracts workers to surround and tend to the queen. It is spread around the colony; workers react quickly to its absence or decline.

queenright

A queenright colony is one that has a laying queen.

race

See subspecies.

reproductive division of labor

Reproductive division of labor is the separation in reproduction among individuals in a colony. In honey bee colonies, the queen caste specializes in reproduction, while the worker caste specialize in non-reproductive duties such as foraging and brood care. Reproductive division of labor is one of the three hallmarks of eusociality.

resin

Resins are sticky substances secreted by plants and collected by bees to make propolis.

retinue

The queen's retinue is the group of young worker bees that surround and care for her.

round dance

A round dance is in fact a mistaken term: waggle dances communicating sources of food that are very close to the hive are so short in duration that they look like a different sort of dance, once called a round dance. We now know that these dances are just a short waggle dance.

royal jelly

Royal jelly is a very nutritious food source secreted by the hypopharyngeal glands and mandibular glands in nurse bees' heads. It is milky in consistency and contains water, sugars, salts, fatty acids and other lipids, vitamins, amino acids, and proteins. Developing queens receive royal jelly during their entire larval phase, whereas developing workers are fed royal jelly for only the first 2-3 days.

scout bee

Scout bees are specialized foragers that search for resources like nectar or pollen, or new nest sites for a swarm.

sex pheromone

Sex pheromones are released by organisms to attract members of the opposite sex to mate with them. In honey bees, queen pheromone is a sex pheromone that attracts drones.

social immunity

Social immunity is collective social behavior that aims to reduce, control, or eliminate parasites and pathogens. Social immunity encompasses a variety of behaviors such as hygienic behavior, grooming, removing dead bodies from the nest, and creating a social fever.

spermatheca

The spermatheca is a sac in a female insect's abdomen that stores sperm after mating. Queens have a spermatheca. Worker bees also have a spermatheca, but relative to the queen's, theirs are tiny and cannot store sperm.

species

A species is a group of organisms that is capable of mating and producing fertile offspring. Species are the basic rank in the classification of life. Species names are always italicized. Honey bees are the species *Apis mellifera*.

spiracle(s)

Spiracles are small holes along the thorax and abdomen of a honey bee that allow respiration to take place. They are the openings where the tracheae exit the body.

stock

A stock is a group of honey bees bred for a specific combination of traits.

subspecies

A subspecies (or race) is a genetically distinct (and often geographically isolated) population of organisms within one species. Subspecies have differences in physical and behavioral traits, but are able to mate with each other, where their offspring are called 'hybrids'. A subspecies name is designated by the third part of a scientific name. For example, the Carniolan bee is called *Apis mellifera carnica*, where *carnica* is the subspecies/race.

sucking pump

The sucking pump is a muscle-lined sac within the mouth of the honey bee that allows it to suck up liquids through its proboscis. The pump also allows for nectar to be regurgitated for sharing with nestmates or storing in cells to make honey.

summer bees

Summer bees are the bees that are born and live during the active, warm season. They are slightly physiologically different from winter bees, which are produced within the colony from late summer to late fall and survive the winter.

supersede

(noun: supersedure)

Supersedure is the natural replacement of an old or failing queen with a younger queen by the workers in the colony.

supersedure cell

Supersedure cells are a specific type of queen cell that worker bees build while their queen is still present in the colony, with the intention of superseding her. These queen cells are distinct from emergency queen cells.

superorganism

A superorganism is a group of related, social individuals working together to effectively cooperate as one large organism.

swarm

noun: A swarm is a cluster of bees that has left its old nest, including the queen of the original colony, that has temporarily settled in the open before locating a new home.
verb: Swarming is the name given to the act of these bees leaving the nest. This process is colony-level reproduction, as only a fraction of bees leave the original nest and the colony effectively divides itself into two (or more) colonies.

swarm cell

Swarm cells are a specific type of queen cell that worker bees build while their queen is still present in the colony. The colony is then prepared to swarm, and the swarmed queen will be naturally replaced with a new queen from a swarm cell.

thermoregulation

Active and passive thermoregulation keeps the colony (and specifically the brood nest) at a constant temperature. This allows the brood to develop properly and the colony to survive weather extremes in a wide range of climates around the world.

thorax

The thorax is the middle segment of an insect's body, located between the head and the abdomen.

trachea

(plural: tracheae)
Tracheae are a series of tubes through which honey bees breathe. They function much like lungs, carrying oxygen to muscles and organs and carrying carbon dioxide away from them. Tracheae exit the body through spiracles.

tremble dance

The tremble dance is a dance performed by foragers to recruit bees to unload nectar from incoming foragers. If a forager that has found an excellent nectar source cannot find a receiver bee to unload her when she returns to the hive, she will perform a tremble dance instead of a waggle dance.

trophallaxis

Trophallaxis is the transfer or exchange of fluids or food mouth-to-mouth (or, proboscis-to-proboscis). Trophallaxis allows bees to feed queens and larvae, unload nectar or water from foraging trips, share pheromones, and more.

ventriculus

The honey bee ventriculus is analogous to a human stomach. It is where food is digested and nutrients are absorbed in the midgut.

vitellogenin

Vitellogenin is a protein found in bees' ovaries (and other tissues) that is important for making eggs. It also has functions in producing royal jelly, caste differentiation, and worker polyethism.

waggle dance

The waggle dance is a dance performed by foragers and scout bees to communicate information about the location of sources of food/resources and nest sites. The pattern indicates direction and distance; the number of times the dance is repeated indicates the quality of the source or nest site. It is a famous example of honey bee communication.

wax

Wax is the malleable substance that honey bees secrete and use to build comb in their nest. House bees producing wax through their wax glands must consume a lot of honey to produce wax. Wax absorbs lipophilic (fat-loving) substances.

wax gland

Wax glands are on the underside of worker bees' abdomens. These glands are most active before the worker transitions to foraging.

wax mirrors

When honey bees produce wax, they excrete liquid wax from their wax glands and deposit it onto plates called wax mirrors, where it hardens into scales or flakes as it dries. The bees then scrape off the hardened wax scales, chew and moisten it until it is the right consistency, and transfer it to the comb.

winter bees

Winter bees are produced within a colony from late summer to late fall to survive the winter. Their physiology differs from that of summer bees insofar as they have lower levels of juvenile hormone, and larger hypopharyngeal glands, and larger and more numerous fat bodies than summer bees.

winter cluster

See cluster.

worker

A worker is a female bee that is not the queen of the colony. The vast majority of bees in a healthy colony are workers, offspring of the queen and the drones that she has mated with. Workers perform many tasks in the colony (see polyethism), but usually refrain from reproduction.

venom

Bee venom is the toxic liquid that is pumped into whatever organism a worker or queen stings. It has over 50 components, including a protein that stimulates the release of histamine in animals (including humans) and other chemicals that together can trigger an allergic response. Barbs in the worker bee's stinger anchor the stinger so that it can continue to pump venom even after the bee disengages or dies. Venom is not the same as the alarm pheromone that is also released to attract and stimulate other bees to sting.